

MENDSPEECH REVISED LEARNING SYSTEM

Week 7: TTS, Speaker Preservation, and Boundary Matched Reconstruction

Build MendSpeech V1 as a cascaded selective repair baseline with explicit seam diagnostics.

Six focused build days plus one consolidation day. Each day ends with named artifacts or measured evidence.

Week milestone

Build MendSpeech V1 as a cascaded selective repair baseline with explicit seam diagnostics.

Week map

Day	Focus	Minimum evidence	Compute
Day 43	TTS system anatomy	• Compare several sentences with punctuation and pacing changes.	Modal L4
Day 44	FastSpeech style duration and prosody	• Change speaking rate or duration settings and measure generated length.	Modal L4
Day 45	Vocoder realism and acoustic boundary diagnostics	• Measure inference speed and real time factor. • Create intentionally mismatched generated spans and verify that the boundary diagnostics flag obvious loudness or spectral discontinuities.	Modal L4
Day 46	VITS and end to end synthesis	• Create a listening sheet with randomized sample order.	Modal L4
Day 47	Speaker representation and preservation	• Compare full resynthesis with short span reconstruction for speaker similarity.	Modal L4
Day 48	Selective reconstruction with boundary matched stitching	• Compare full utterance TTS, naive selective repair, and boundary matched selective repair. • Measure preservation percentage, latency, energy discontinuity, and speaker similarity proxy. • Run a small blinded seam audibility check with randomized sample order.	Modal L4 plus local CPU for stitching
Day 49	Week 7 MendSpeech V1 cascaded repair milestone	• Run at least ten cases, including deliberate false repair, missed repair, seam artifacts, and one case where the policy abstains. • Compare naive stitching and boundary matched stitching on the same repaired spans.	Modal L4

Reference spine

- FastSpeech 2 paper
- HiFi GAN paper
- VITS paper
- DSP references for energy matching and equal power crossfades

TTS system anatomy

LEARN

- Text or phoneme representation.
- Acoustic model.
- Mel spectrogram or latent representation.
- Vocoder.
- Speaker conditioning.
- Prosody.

BUILD IN MENDSPEECH

- Run a pretrained TTS system on controlled text.
- Save generated waveform and intermediate representations if exposed.

EXPERIMENT AND MEASURE

- Compare several sentences with punctuation and pacing changes.

REQUIRED OUTPUT

- src/tts/baseline.py
- results/day43_tts_samples/
- docs/tts_pipeline.md

Completion check: You can explain how text becomes waveform and where speaker identity can enter.

Study method

25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

REFERENCE PRIORITY

- FastSpeech 2 paper
- HiFi GAN paper
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FastSpeech style duration and prosody

LEARN

- Duration prediction.
- Pitch and energy predictors.
- Parallel generation intuition.

BUILD IN MENDSPEECH

- Study FastSpeech 2 architecture and inspect an implementation.
- Extract or visualize duration, pitch, or energy controls if available.

EXPERIMENT AND MEASURE

- Change speaking rate or duration settings and measure generated length.

REQUIRED OUTPUT

- docs/day44_fastspeech2.md
- results/day44_prosody_samples/

Completion check: You understand why duration matters when replacing only a short span.

Study method

25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

REFERENCE PRIORITY

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Vocoder realism and acoustic boundary diagnostics

LEARN

- Mel to waveform generation.
- HiFi GAN style generator and discriminator intuition.
- Phase, bandwidth, and vocoder artifacts.
- Short time energy, local loudness, spectral balance, and room tone as boundary signals.

BUILD IN MENDESPEECH

- Run a neural vocoder or inspect the one used by the selected TTS stack.
- Add boundary diagnostics that measure short time energy and simple spectral statistics before and after a candidate repair span.
- Save a local room tone estimate where possible.

EXPERIMENT AND MEASURE

- Measure inference speed and real time factor.
- Create intentionally mismatched generated spans and verify that the boundary diagnostics flag obvious loudness or spectral discontinuities.

REQUIRED OUTPUT

- results/day45_vocoder_benchmark.csv
- src/repair/boundary_metrics.py
- docs/vocoder_and_boundary_notes.md

Completion check: You can separate acoustic model errors from vocoder artifacts and quantify at least two causes of an audible seam.

Study method

25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

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VITS and end to end synthesis

LEARN

- Latent variable modeling.
- Normalizing flow intuition.
- Adversarial training.
- End to end waveform generation.

BUILD IN MENDESPREECH

- Run a VITS style pretrained model or study its code path.
- Compare latency and perceived naturalness with your Week 7 baseline.

EXPERIMENT AND MEASURE

- Create a listening sheet with randomized sample order.

REQUIRED OUTPUT

- results/day46_tts_comparison.csv
- results/day46_listening_sheet.md

Completion check: You can explain why different TTS architectures behave differently for selective repair.

Study method

25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

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Speaker representation and preservation

LEARN

- Speaker embeddings.
- Reference conditioned synthesis.
- Speaker similarity as a measurable but imperfect proxy.
- Consent and voice identity boundaries.

BUILD IN MENDSPEECH

- Choose a speaker conditioned or reference conditioned path that is legally and ethically appropriate for your own or consented samples.
- Compute speaker embeddings before and after synthesis if tooling is available.

EXPERIMENT AND MEASURE

- Compare full resynthesis with short span reconstruction for speaker similarity.

REQUIRED OUTPUT

- src/tts/speaker_conditioning.py
- results/day47_speaker_similarity.csv
- docs/voice_use_policy.md

Completion check: You can discuss speaker similarity measurements and their limitations without claiming identity preservation from listening alone.

Study method

25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

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Selective reconstruction with boundary matched stitching

LEARN

- Repair span text selection.
- Timing constraints and duration control.
- Boundary padding and silence handling.
- Short time energy matching and local loudness matching.
- Linear versus equal power crossfades.
- Spectral and room tone mismatch.
- Why ASR to text to TTS can lose pitch, emotion, breathing, and coarticulation.

BUILD IN MENDSPEECH

- Take a known damaged interval and synthesize only its transcript span.
- Match generated duration to the target interval without changing untouched speech.
- Match local energy before stitching and implement both linear and equal power crossfades.
- Add optional room tone under the regenerated span when the original context supports it.
- Log preserved samples, reconstructed samples, boundary length, and all matching parameters.

EXPERIMENT AND MEASURE

- Compare full utterance TTS, naive selective repair, and boundary matched selective repair.
- Measure preservation percentage, latency, energy discontinuity, and speaker similarity proxy.
- Run a small blinded seam audibility check with randomized sample order.

REQUIRED OUTPUT

- `src/repair/reconstruct.py`
- `src/repair/stitch.py`
- `src/repair/boundary_metrics.py`
- `results/day48_selective_samples/`
- `results/day48_seam_ablation.csv`

Completion check: The final audio keeps most original samples, replaces only a targeted interval, and shows measurably smoother boundaries than naive stitching.

Study method

25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

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Week 7 MendSpeech V1 cascaded repair milestone

LEARN

- Review TTS, duration, vocoder behavior, speaker conditioning, boundary matching, and information lost through the text bottleneck.
- Treat the cascaded path as a real time baseline, not as the final frontier of speech restoration.

BUILD IN MENDSPEECH

- Pipeline: damaged audio to streaming ASR to uncertain span to policy decision to speaker conditioned reconstruction to boundary matched waveform.
- Show preserved and reconstructed intervals with distinct visualization.
- Add a V1 label in results so the Week 8 direct audio repair comparison is explicit.

EXPERIMENT AND MEASURE

- Run at least ten cases, including deliberate false repair, missed repair, seam artifacts, and one case where the policy abstains.
- Compare naive stitching and boundary matched stitching on the same repaired spans.

REQUIRED OUTPUT

- app/mendspeech_v4_cascaded.py
- demos/week7_before_after/
- results/week7_stitching_ablation.csv
- reports/week7_cascaded_repair.md

Completion check: MendSpeech V1 is a measured cascaded baseline whose strengths and prosody or seam limitations are documented rather than hidden.

Study method

25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

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